

Employment

Senior Systems Engineer at Cloudflare

December 2023 - Present

- Implemented full product functionality for MySQL, supporting entirely new user base and substantially growing TAM for the product. [See article.](#) (Rust, Go, TypeScript)
- Re-architected Hyperdrive's core query processing and caching logic to facilitate local-first caching, reducing cache hit latency by 95%. (Rust)
- Developed placement subsystem that optimally located Cloudflare resources near customer origin databases, reducing query latency by up to 80%. [See article.](#) (Rust, Go, TypeScript)
- Designed and implemented support for Postgres prepared statements in distributed connection pooler, unblocking key driver in the JavaScript ecosystem. [See article.](#) (Rust)
- Developed integration with Cloudflare Tunnels to enable full product compatibility with private networks. Developed custom websocket implementation to facilitate this. [See article.](#) (Rust)

Software Engineer at VMware

June 2022 - November 2023

- Developed key features within backup and restore utilities for Greenplum database. Notably includes parallel metadata restore, reducing latency of schema restore by up to 200x in customer systems. [See article.](#) (Go)
- Architected and implemented from-scratch rewrite of backup manager utilities, reducing runtime latency by approximately 800x on very large customer deployments. (Go, sqlite)
- Designed and developed new SQL command (ALTER TABLE REPACK) to improve compression and scan performance on very large customer datasets. Observed 50% reduction in disk space usage in tests of large installs. (C)
- Implemented several key features in core Greenplum server codebase, to help drive to release of GPDB7. Examples include autovacuum implementation and parallelized system monitoring views. [See video tutorial.](#) (C)
- [Dedicated GitHub profile for my open source contributions at VMware](#)

Software Engineer at Network Perception

April 2021 - June 2022

- Profiled and improved performance of core backend, inference, and parser code. Resulted in approximately 100x reduction in operation latency (hours to seconds). (Python, Redis)
- Designed and implemented a full rebuild of the product's integrated analytics engine. (Python, JavaScript)
- Developed parser and inference integration for various networking software output formats, including Cisco ASA, Fortilink Fortiguard, Google Cloud Platform, AWS Networking, Nessus, nmap, and libpcap. (Python)

Senior Data Scientist & Team Lead at OptionCare Health

July 2019 - April 2021

- Developed software package for analysis of patient medication adherence. Authored internal whitepaper showing company excellence relative to industry standards, submitted to conference. (Python, SQL)
- Developed schema and ETL queries for analytic data mart, leveraging existing primary data sources into an enriched platform for rapid analysis. (SQL)

Senior Analyst & Engineering Manager at OptumRx

January 2017 - June 2019

- Developed statistical programs deployed to visualization layer hosted on department website. Owned every step of multiple projects using this workflow, from requirements to deployment. (SAS, Tableau, Azure)
- Scoped, architected, developed, and deployed a new algorithmic approach to optimally categorize products by price to minimize cost to the company and maximize patient satisfaction. This was successfully productized by the company. (SAS, SQL).

Notable Technical Projects

- OSS Medication Adherence Package (2020). Developed and published package for calculating medication adherence. Used pandas, NumPy, and multiprocessing. Contracted as a consultant on its use and validation for multiple private healthcare entities. Ported to rust with clap and rayon. Discussed [here](#). (Python, Rust)
- Formulary Optimization System (2019). Developed and deployed a program to analyze medication usage across 65 million patients using customized algorithm and grid search. Productized by employer. (SAS, SQL)

Education

Georgia Institute of Technology	Master of Science - Computer Science
University of Wisconsin (Stevens Point)	Master of Science - Fisheries Science
University of Illinois (Champaign-Urbana)	Bachelor of Science - Natural Resources